

Machine-checked proofs of ac-LEEM Fourier-optics identities

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import LeemF0

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Abstract

This note is a literate Lean 4 rendering of the formal proofs in the **LeemF0** library: the appendix identities of Yu, Lau and Altman, *Ultramicroscopy* **200** (2019) 160–168, for Fourier-optics (FO) image formation in aberration-corrected LEEM. Each section states the analytic claim, points at the Lean name, and replays the proof script that the library uses.

The PDF is produced by [Alectryon](#) from this file (`docs/alectryon/LeemF0Proofs.lean`). Alectryon extracts the reStructuredText prose delimited by `/-| ... |-/` and typesets the surrounding Lean. The companion handwritten analytic note is `docs/proofs/leemfo_proofs.pdf`; equation numbers and the modelling hypotheses live in `docs/FORMALIZATION.md`.

Conventions

- χ is in **waves**, with $W = e^{2\pi i\chi}$.
- Lean cannot use λ as an identifier, so the wavelength is `lam`.
- The 1D theory uses signed scalars q, q' (products, not magnitudes).
- Truncations used by the paper (frozen aperture, first-order tilt, dropped mixed $k\varepsilon$) are modelling hypotheses, not Lean theorems.

open Complex Real Filter Asymptotics Topology

open scoped ComplexConjugate Topology

open LEEM

namespace LeemF0Proofs

Instrument model

The structure `LEEM` packages one imaging condition. Spatial and chromatic paths are the spatial-frequency forms of the paper, with powers of `lam` restored so that q is a spatial frequency:

$$\chi_S(q, \Delta z) = \frac{1}{4}C_3\lambda^3q^4 + \frac{1}{6}C_5\lambda^5q^6 + \frac{1}{2}\Delta z\lambda q^2,$$

$$\chi_C(q, \varepsilon) = \frac{1}{2}C_C\lambda(\varepsilon/E)q^2 + \frac{1}{2}C_{CC}\lambda(\varepsilon/E)^2q^2 + \frac{1}{4}C_{3C}\lambda^3(\varepsilon/E)q^4.$$

The coherent kernel R_0 (paper Eq. (2)) is the product of apertures and transfer factors. After the paper's first-order / no-mixed-term approximation, the working FO kernel is $R_{\text{FO}} = R_0(q, q', \Delta z, 0) E_S E_{C, \text{tot}}$.

#check LEEM

#check LEEM.chiS

#check LEEM.chiC

#check LEEM.R0

#check LEEM.aS

#check LEEM.b1

#check LEEM.b2

The phase identity $W_S(q)W_S(q')^* = \exp(2\pi i(\chi_S(q) - \chi_S(q')))$ is `LEEM.waveS_mul_conj`. It is `cexp`-algebra: conjugating flips the sign of $2\pi i\chi$.

```
example (p : LEEM) (q q' Δz : ℝ) :
  p.waveS q Δz * conj (p.waveS q' Δz)
    = cexp (2 * π * I * ((p.chiS q Δz - p.chiS q' Δz : ℝ) : ℂ)) :=
  p.waveS_mul_conj q q' Δz
```

Gaussian calculus

The paper's Gaussians are specified by FWHM. With $\Delta = 2\sqrt{2\ln 2}\sigma$ one has $f(\Delta/2) = f(0)/2$, and inverting gives $\sigma^2 = \Delta^2/(8\ln 2)$. The FWHM packaging of the spatial envelope uses $\kappa = \pi^2 q_{\text{ill}}^2/(4\ln 2)$, which is exactly $2\pi^2\sigma_{\text{ill}}^2$.

```
example {σ : ℝ} (hσ : 0 < σ) :
  rexp (-(fwhmFactor * σ) / 2) ^ 2 / (2 * σ ^ 2) = 1 / 2 := by
  have hhalf : (fwhmFactor * σ) / 2 = Real.sqrt (2 * Real.log 2) * σ := by
  unfold fwhmFactor; ring
  have hsq : ((fwhmFactor * σ) / 2) ^ 2 / (2 * σ ^ 2) = Real.log 2 := by
  rw [hhalf, mul_pow, Real.sq_sqrt (mul_nonneg two_pos.le log_two_pos.le)]
  field_simp [ne_of_gt hσ]
  have hneg : -((fwhmFactor * σ) / 2) ^ 2 / (2 * σ ^ 2) = -Real.log 2 := by
  rw [neg_div, hsq]
  rw [hneg, Real.exp_neg, Real.exp_log two_pos]
  ring
```

```
example {σ Δ : ℝ} (hσ : 0 < σ) (hΔ : Δ = fwhmFactor * σ) :
  σ ^ 2 = Δ ^ 2 / (8 * Real.log 2) :=
  variance_from_fwhm hσ hΔ
```

```
example {σ qIll : ℝ} (hσ : 0 < σ) (h : qIll = fwhmFactor * σ) :
  2 * π ^ 2 * σ ^ 2 = kappaIll qIll :=
  kappaIll_eq_sigma hσ h
```

Characteristic function

Appendix A1 is the Fourier transform of a centred Gaussian. Lean proves

$$\int_{\mathbb{R}} s(k) e^{2\pi i a k} dk = e^{-2\pi^2 \sigma^2 a^2}$$

as `charFun_source1D`. The proof rewrites the integrand as a quadratic exponential, applies Mathlib's `integral_cexp_quadratic`, cancels the normalisation against $\sqrt{\pi/|B|}$, and uses $i^2 = -1$ in the completed-square exponent.

```
example {σ a : ℝ} (hσ : 0 < σ) :
  ∫ k : ℝ, (gaussian1D σ k : ℂ) * cexp ((2 * π * I) * a * k)
  = cexp (-(2 : ℂ) * π ^ 2 * σ ^ 2 * a ^ 2) :=
  charFun_source1D hσ
```

The 2D isotropic source is a product of 1D Gaussians, so the 2D characteristic function is the 1D result iterated (`charFun_source2D`). The exponent becomes $-2\pi^2\sigma^2(a_x^2 + a_y^2)$.

Aberration identities

`LEEM.hasDerivAt_chiS` differentiates the polynomial χ_S . The illumination-tilt factor is then exactly the difference of derivatives, `LEEM.aS_eq_deriv_sub`. Paper Eq. (3) is the little-o form of that derivative (`LEEM.chiS_taylor`).

```
example (p : LEEM) (q q' Δz : ℝ) :
  p.aS q q' Δz = deriv (p.chiS · Δz) q - deriv (p.chiS · Δz) q' := by
```

```
simp [LEEM.aS, p.deriv_chiS]
ring
```

```
example (p : LEEM) (Δz q : ℝ) :
  (fun k : ℝ => p.chiS (q + k) Δz - p.chiS q Δz
    - k * (p.C3 * p.lam ^ 3 * q ^ 3 + p.C5 * p.lam ^ 5 * q ^ 5
      + Δz * p.lam * q))
  =o[nhds 0] id :=
  p.chiS_taylor Δz q
```

The chromatic path difference is not an approximation: it factors exactly as $b_1\varepsilon + b_2\varepsilon^2$ once $E \neq 0$ (LEEM.chiC_sub). On the diagonal $q = q'$ one has $a = b_1 = b_2 = 0$.

```
example (p : LEEM) (hp : p.IsPhysical) (q q' ε : ℝ) :
  p.chiC q ε - p.chiC q' ε = p.b1 q q' * ε + p.b2 q q' * ε ^ 2 := by
  unfold LEEM.chiC LEEM.b1 LEEM.b2 LEEM.IsPhysical at *
  have hE : p.E ≠ 0 := ne_of_gt hp.2
  field_simp [hE]
  ring
```

```
example (p : LEEM) (q Δz : ℝ) :
  p.aS q q Δz = 0 ∧ p.b1 q q = 0 ∧ p.b2 q q = 0 :=
  ⟨p.aS_zero_diag q Δz, p.b1_zero_diag q, p.b2_zero_diag q⟩
```

Spatial envelope (paper Eq. (5))

The spatial envelope is the source characteristic function evaluated at $a = \partial_q \chi_S(q) - \partial_q \chi_S(q')$:

$$E_S(q, q', \Delta z) = \int s(k) e^{2\pi i a k} dk = e^{-2\pi^2 \sigma_{\text{in}}^2 a^2} = e^{-\kappa a^2}.$$

spatialEnvelopeIntegral_eq_closed is the first equality (instantiate charFun_source1D); spatialEnvelopeClosed_eq_fwhm rewrites $2\pi^2 \sigma^2$ as κ ; chaining them is spatialEnvelope_eq_fwhm.

```
example (p : LEEM) (h : 0 < p.qIll) (q q' Δz : ℝ) :
  p.spatialEnvelopeIntegral q q' Δz = p.spatialEnvelopeClosed q q' Δz := by
  unfold LEEM.spatialEnvelopeIntegral LEEM.spatialEnvelopeClosed
  rw [charFun_source1D (p.sigmaIll_pos h)]
  congr 1
  push_cast
  ring
```

```
example (p : LEEM) (h : 0 < p.qIll) (q q' Δz : ℝ) :
  p.spatialEnvelopeClosed q q' Δz = p.spatialEnvelopeFWHM q q' Δz := by
  unfold LEEM.spatialEnvelopeClosed LEEM.spatialEnvelopeFWHM
  have hk := kappaIll_eq_sigma (p.sigmaIll_pos h) (p.sigmaIll_eq h)
  congr 1
  simp [← hk]
```

```
example (p : LEEM) (h : 0 < p.qIll) (q q' Δz : ℝ) :
  p.spatialEnvelopeIntegral q q' Δz = p.spatialEnvelopeFWHM q q' Δz :=
  p.spatialEnvelope_eq_fwhm h q q' Δz
```

The closed form is real (`LEEM.spatialEnvelopeClosed_real`) and symmetric in (q, q') because $a(q, q') = -a(q', q)$. On the diagonal $a = 0$ so $E_S = 1$.

Chromatic envelope (paper Eqs. (6a)–(6b))

Energy spread contributes both a linear phase $b_1\varepsilon$ and a quadratic phase $b_2\varepsilon^2$. Completing the square in the complex Gaussian integral produces

$$E_{CC} = (1 - 4\pi i b_2 \sigma_E^2)^{-1/2}, \quad E_{C,\text{tot}} = E_{CC} \exp(-2\pi^2 \sigma_E^2 b_1^2 E_{CC}^2).$$

Lean names: `ecc` and `chromaticEnvelopeClosed`. The identity `chromaticEnvelopeIntegral_eq_closed` matches the integral to that closed form: rewrite as a quadratic exponential (`chromatic_integral_eq_quadratic`, using Mathlib `integral_cexp_quadratic`) and identify the prefactor with `ecc`. If $b_2 = 0$ (nac), `ecc = 1` and the formula collapses to the ordinary characteristic function.

```
#check ecc
```

```
#check chromaticEnvelopeClosed
```

```
example {σ b1 : ℝ} :
  chromaticEnvelopeClosed σ b1 0
  = cexp (-(2 : ℂ) * π ^ 2 * σ ^ 2 * b1 ^ 2) :=
  chromaticEnvelopeClosed_nac
```

```
example (p : LEEM) (h : 0 < p.ΔE) (q q' : ℝ) :
  p.chromaticEnvelopeIntegral q q'
  = chromaticEnvelopeClosed p.sigmaE (p.b1 q q') (p.b2 q q') :=
  p.chromaticEnvelopeIntegral_eq_closed h q q'
```

Polar form of E_{CC}

Write $1 - 4\pi i b_2 \sigma_E^2 = 1 - iy$ with $y = 4\pi b_2 \sigma_E^2$. Then $\arg(1 - iy) = -\arctan y$ (real part is positive) and

$$(1 - iy)^{-1/2} = (1 + y^2)^{-1/4} e^{i(\arctan y)/2}.$$

That is `ecc_eq_polar`, using `arg_one_sub_I` and `ecc_polar`.

```
example (σ b2 : ℝ) :
  ecc σ b2
  = Complex.ofReal
    ((1 + (4 * Real.pi * b2 * σ ^ 2) ^ 2) ^ (-(1 / 4 : ℝ)))
    * cexp (I * (1 / 2) * arctan (4 * Real.pi * b2 * σ ^ 2)) :=
  ecc_eq_polar σ b2
```

CTF as the $q' = 0$ slice

`R_F0` is the working bilinear kernel; `R_CTF` is the separable product of one-argument envelopes (printed Eqs. (7a)–(7b), **no** extra conjugate on the chromatic factors). Hermiticity of R_0 is `LEEM.R0_hermitian`: conjugating swaps (q, q') because apertures are real.

On $q' = 0$ the joint envelopes reduce to the CTF product (`LEEM.R_F0_eq_R_CTF_axis`): the spatial diagonal is 1, and $b_1(q, 0)$, $b_2(q, 0)$ together with the nac collapse of the primed chromatic factor match `spatialCTF` and `chromaticCTF`.

```
#check LEEM.R_F0
#check LEEM.R_CTF
```

```
example (p : LEEM) (q q' Δz ε : ℝ) :
  conj (p.R0 q q' Δz ε) = p.R0 q' q Δz ε := by
  simp only [LEEM.R0, map_mul, conj_ofReal, conj_conj]
  ac_rfl
```

```
example (p : LEEM) (q Δz : ℝ) :
  p.R_F0 q 0 Δz = p.R_CTF q 0 Δz := by
  unfold LEEM.R_F0 LEEM.R_CTF LEEM.spatialCTF LEEM.chromaticCTF
  rw [p.spatialEnvelopeClosed_diag 0 Δz, p.b1_zero_diag, p.b2_zero_diag,
    chromaticEnvelopeClosed_nac]
  simp
```

Envelope ratios

The comparison factors are the ratios of axis envelopes to joint envelopes:

$$\Gamma_S = \frac{E_S(q, 0) E_S(q', 0)}{E_S(q, q')}, \quad \Gamma_C = \frac{E_C(q, 0) E_C(q', 0)}{E_C(q, q')}.$$

Gaussian algebra gives $\Gamma_S = \exp(-4\pi^2 \sigma_{\text{III}}^2 uv)$ with $u = \chi'(q)$, $v = \chi'(q')$ (LEEM.gammaS_eq_dot), or equivalently $\exp(-2\kappa uv)$ (LEEM.gammaS_eq_kappa). Under AC ($C_3 = 0$) the derivatives keep only C_5 and defocus (LEEM.gammaS_ac).

```
example (p : LEEM) (q q' Δz : ℝ) :
  p.gammaS q q' Δz
  = cexp (-(4 : ℂ) * π ^ 2 * p.sigmaIll ^ 2
    * deriv (p.chiS · Δz) q * deriv (p.chiS · Δz) q') :=
  p.gammaS_eq_dot q q' Δz
```

```
example (p : LEEM) (h : 0 < p.qIll) (q q' Δz : ℝ) :
  p.gammaS q q' Δz
  = cexp (-(2 : ℂ) * kappaIll p.qIll
    * deriv (p.chiS · Δz) q * deriv (p.chiS · Δz) q') :=
  p.gammaS_eq_kappa h q q' Δz
```

```
example (p : LEEM) (hAC : p.IsAC) (h : 0 < p.qIll) (q q' Δz : ℝ) :
  p.gammaS q q' Δz
  = cexp (-(2 : ℂ) * kappaIll p.qIll
    * (p.C5 * p.lam ^ 5 * q ^ 5 + Δz * p.lam * q)
    * (p.C5 * p.lam ^ 5 * q' ^ 5 + Δz * p.lam * q')) :=
  p.gammaS_ac hAC h q q' Δz
```

Multiplying FO by the two ratios rebuilds CTF, because E_S is real (so the spatial conjugate is redundant): LEEM.R_CTF_eq_R_F0_mul_gamma. Perfect coherence ($q_{\text{III}} = \Delta E = 0$) forces both ratios to 1, hence $R_{\text{CTF}} = R_{\text{FO}}$.

```
example (p : LEEM) (q q' Δz : ℝ) :
  p.R_CTF q q' Δz = p.R_F0 q q' Δz * p.gammaC q q' * p.gammaS q q' Δz := by
  have hEs : p.spatialEnvelopeClosed q q' Δz ≠ 0 := Complex.exp_ne_zero _
```

```

have hEc := chromaticEnvelopeClosed_ne_zero p.sigmaE (p.b1 q q') (p.b2 q q')
unfold LEEM.R_CTF LEEM.R_F0 LEEM.gammaC LEEM.gammaS
rw [p.spatialCTF_conj q' Δz]
unfold LEEM.spatialCTF LEEM.chromaticCTF
field_simp [hEs, hEc]

```

```

example (p : LEEM) (h : p.PerfectCoherence) (q q' Δz : ℝ) :
  p.gammaC q q' * p.gammaS q q' Δz = 1 :=
  p.gamma_product_perfect h q q' Δz

```

Jacobi–Anger

A strong phase object $e^{i\varphi \sin \theta}$ has Fourier coefficients equal to Bessel functions. Lean defines `besselJ n φ` as that coefficient on $(0, 2\pi]$, proves $O(1/n^2)$ decay by two integrations by parts (`besselJ_bound`), hence summability, and applies Mathlib’s pointwise Fourier inversion on the circle (`jacobi_anger_on_Ioc`). The identity for every real angle reduces modulo 2π (`jacobi_anger`).

```

#check besselJ
#check jacobi_anger_on_Ioc
#check jacobi_anger

```

```

example (φ θ : ℝ) :
  HasSum (fun n : ℤ => besselJ n φ * cexp (I * n * θ)) (phaseFun φ θ) :=
  jacobi_anger φ θ

```

Lean name map

- Paper $\lambda \rightarrow$ LEEM.lam
- $\chi_S, \chi_C \rightarrow$ LEEM.chiS, LEEM.chiC
- $R_0 \rightarrow$ LEEM.R0
- $a, b_1, b_2 \rightarrow$ LEEM.aS, LEEM.b1, LEEM.b2
- FWHM / $\kappa \rightarrow$ fwhmFactor, kappaIll
- $\int s e^{2\pi i a k} \rightarrow$ charFun_source1D
- Eq. (5) $E_S \rightarrow$ LEEM.spatialEnvelope_eq_fwhm
- Eqs. (6a)–(6b) $\rightarrow$ LEEM.chromaticEnvelopeIntegral_eq_closed
- Polar $E_{CC} \rightarrow$ ecc_eq_polar
- $R_{FO} \rightarrow$ LEEM.R_F0
- $R_{CTF} \rightarrow$ LEEM.R_CTF
- CTF from FO on $q' = 0 \rightarrow$ LEEM.R_F0_eq_R_CTF_axis
- $\Gamma_S = e^{-2\kappa uv} \rightarrow$ LEEM.gammaS_eq_kappa
- $R_{CTF} = R_{FO} \Gamma_C \Gamma_S \rightarrow$ LEEM.R_CTF_eq_R_F0_mul_gamma
- Jacobi–Anger $\rightarrow$ jacobi_anger

Rebuild

From the repository root, with `uv` and TeX Live (LuaLaTeX):

```
make -C docs/alectryon pdf
```

The literate source is this file; the typeset PDF is `docs/alectryon/leemfo_alectryon.pdf`.

`end` LeemF0Proofs